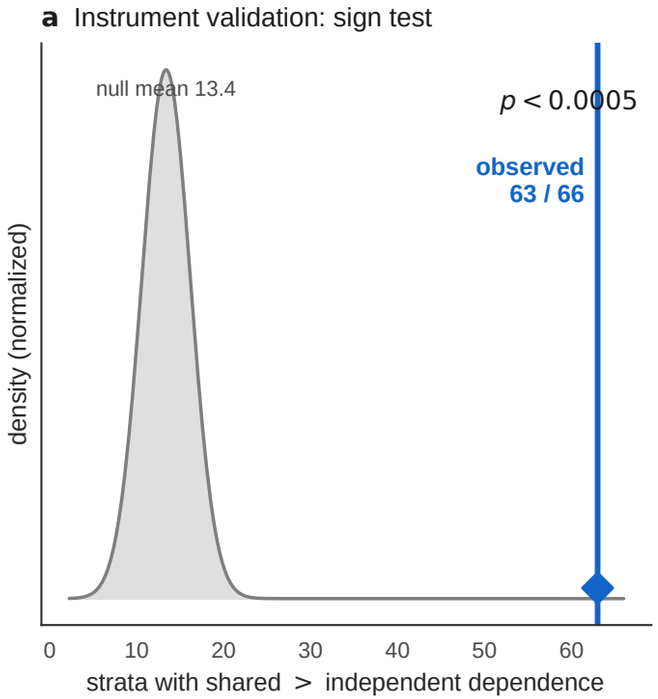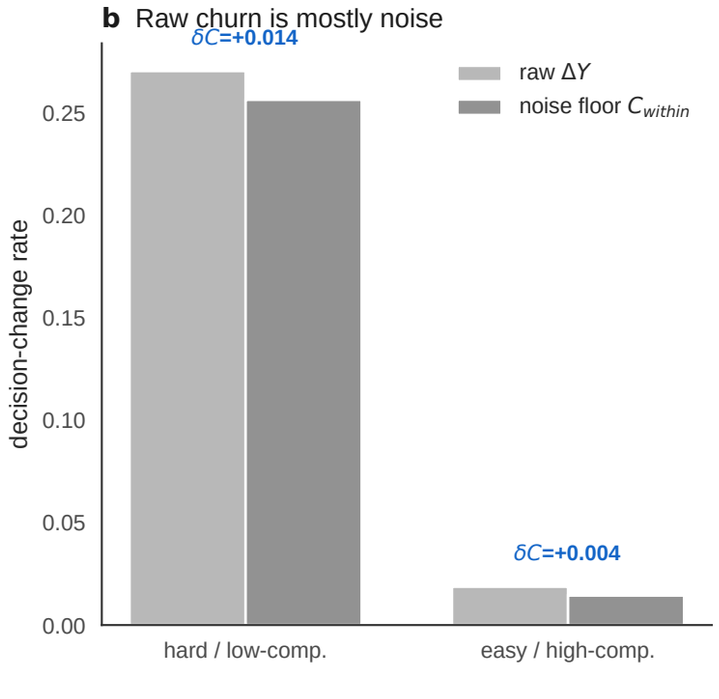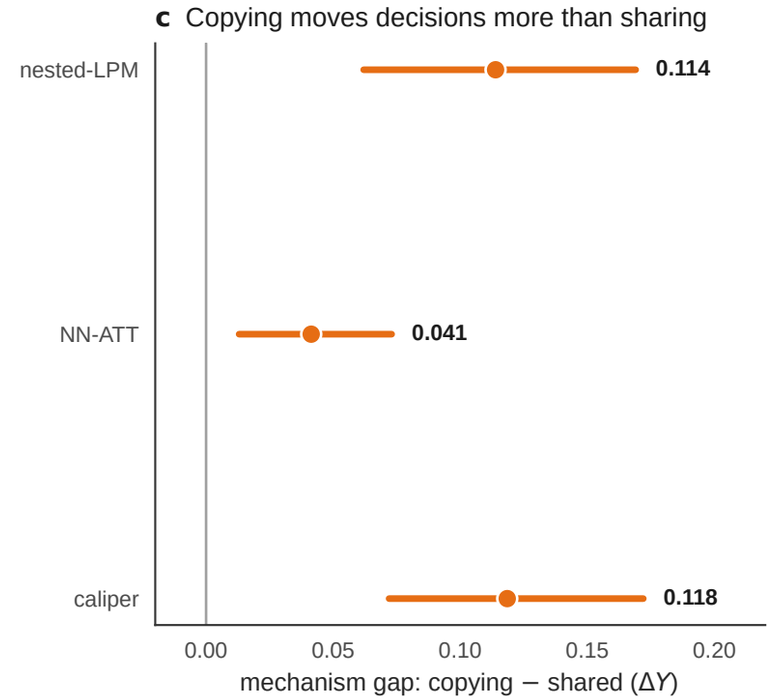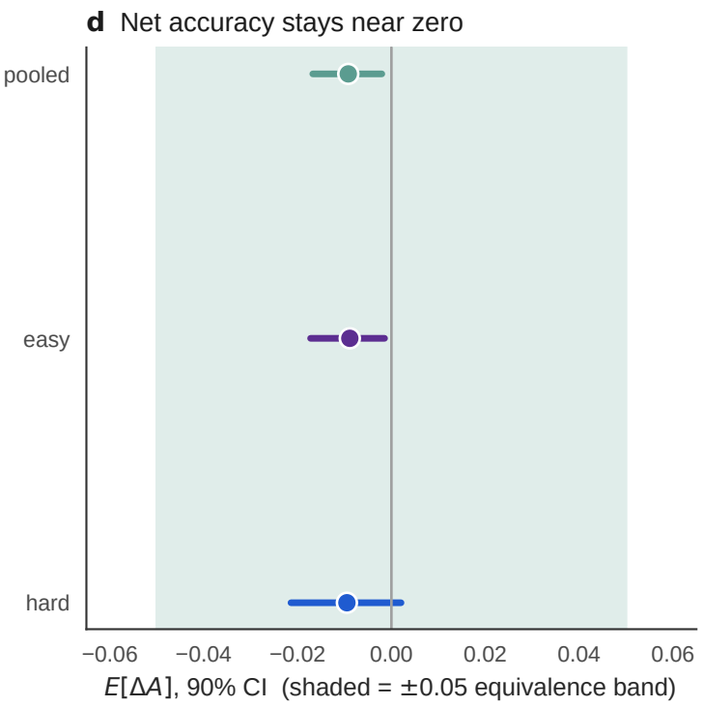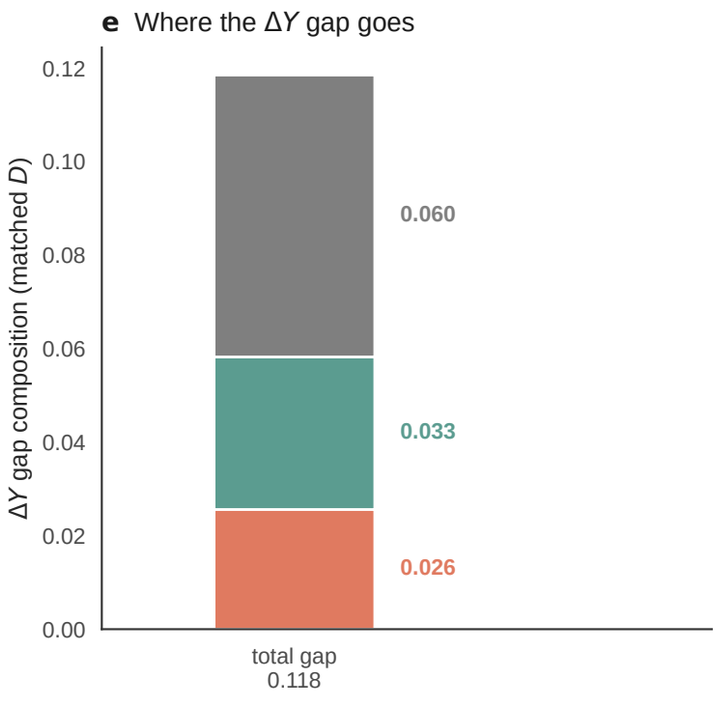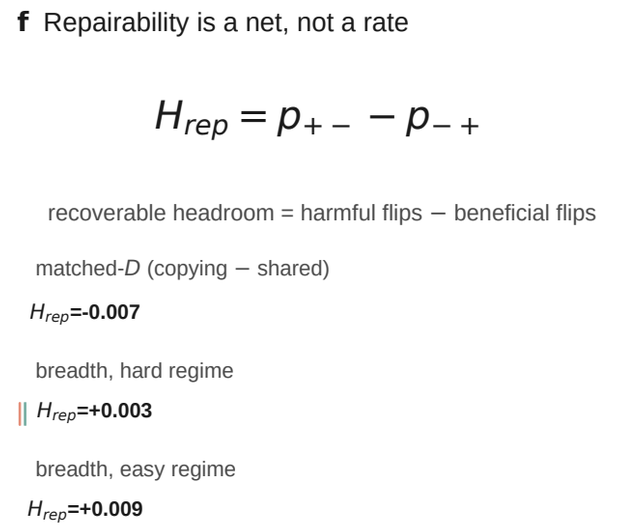Dependence is not enough